

Density-Based TRAFFIC LIGHT CONTROLLER

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Efficient traffic management in urban areas remains a significant challenge as vehicle density grows and infrastructure struggles to keep pace. Conventional traffic lights rely on fixed timers, which often result in unnecessary delays and congestion during low-traffic periods. To address this, an intelligent traffic signal control system has been developed using low-cost, readily available components, including

Arduino Uno and Arduino Nano boards and HC-SR04

ultrasonic sensors. Four ultrasonic sensors continuously monitor vehicle presence across four lanes. The sensor unit (Arduino Uno) collects proximity data and sends it via I2C communication to the control unit (Arduino Nano), which adjusts signal timing in real time.

When a vehicle is detected in a lane, that lane receives a 10-second green light. If no vehicle is present, the green phase is limited to a pre-timed 2 seconds. After the green interval, the signal automatically switches to red, and the system advances to the next lane. The distance readings from each lane are updated on the serial monitor every second.

RGB LEDs indicate the traffic signal status (green/red), and the system logs its control sequence through the serial monitor. The modular, low-cost architecture enables straightforward scaling for additional lanes or future IoT enhancements. Fig. 1 presents a prototype of the

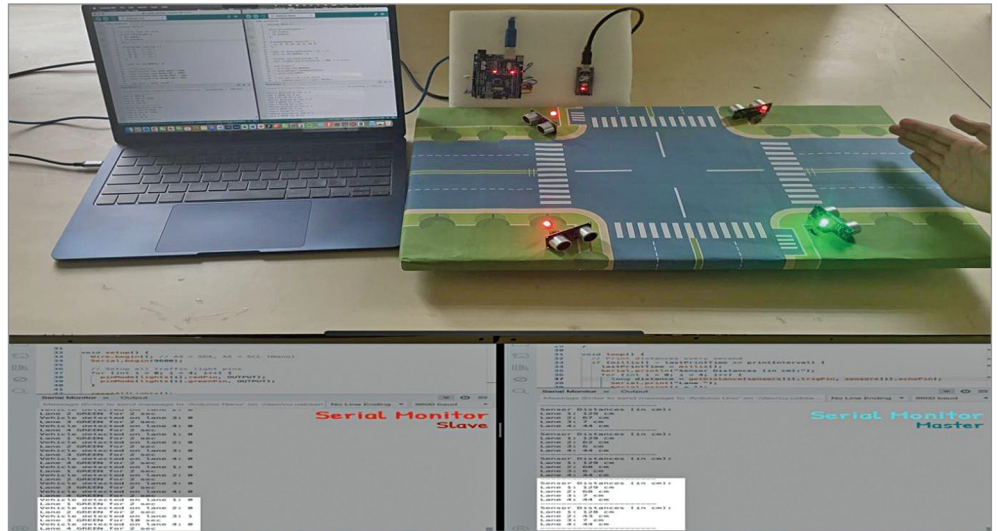


Fig. 1: Prototype of the proposed setup for optimised traffic control

system setup, and Fig. 2 provides the corresponding block diagram.

Two Arduino boards are used in the system. One functions as the sensor device and the other acts as the controller device. Both communicate over the I2C line. The sensor device detects vehicles using ultrasonic sensors, while the controller operates the RGB LED for signal indication. These components form the core of the system design. The required components are listed in Table 1 for the Bill of Materials.

Table 2 summarises the signal behaviour of the smart traffic control system based on real-time vehicle detection. It outlines how the system dynamically adjusts the durations of the green and red LED

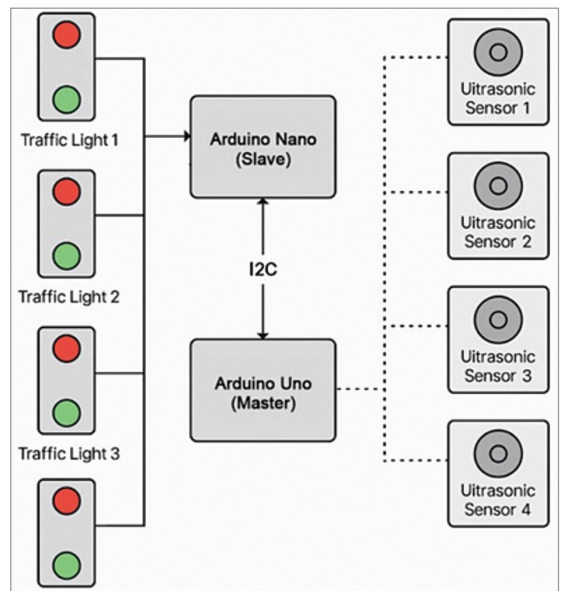


Fig. 2: Block diagram

indicators based on traffic conditions in each lane.

Design, working, and setup

The system has been developed as a dynamic traffic signal control system